

NOVEL FEATURES ARE INVOLVED IN BUILDING OF SPAN

Design of New Bridge Made to Accommodate Conditions Here; Depth of Willamette Is Factor in Final Plan.

FOUNDATION OF ARCH IS UPON SOLID ROCK

Old Towers Are Utilized In Erection of Steel; Three Finishing Processes Used.

(By C. P. Richards, C. E.)

With permanence as the chief consideration, seven types of bridges were examined as to their suitability for this location. A reinforced concrete arch would have met this condition better than any other, but had to be rejected because the depth of the channel and the navigation requirements precluded the use of falsework of the ordinary type; and suspended falsework proved too costly. A steel bridge, whether it be of the suspension, cantilever, simple truss or framed arch type, must be protected from atmospheric corrosion to be classed as truly permanent. Protection by paint is, at best, only temporary and involves heavy maintenance costs. Protection by encasing in concrete is as nearly perfect as could be desired, but, for the usual types of structural steel bridges, involves a prohibitive cost. To overcome this the adopted design was evolved—a bifurcated steel arch, having its ribs of rectangular section with all stiffening members inside and the smooth exterior entirely encased in concrete of a special kind known as "Gunite." Gunite is a mixture of sand and cement shot dry through a hose by compressed air, water being added at the nozzle. It makes a dense, impervious concrete and, on this bridge it is reinforced by a stout wire mesh, previously welded electrically to the sides of the steel ribs. The floor beams and stringers are similarly covered, while the hangers, spandrel columns and other minor parts are encased in poured concrete.

Foundations on Rock

The approaches to the main span are the usual beam and slab type, carried on tapered columns and are of reinforced concrete throughout. Foundations are on solid rock: only one, that of the west main pier, had to be sunk through water. This was done by means of a double walled cofferdam which was pumped dry and the rock bed cleaned of all loose material, concrete then being poured into forms placed on the rock.

The erection of the arch was accomplished by using the anchors, towers and main cables of the old suspension bridge. Each rib consisted of eighteen segments: the first six out from each pier were erected by suspending the upper end on a cable passing over the top of the adjacent tower and thence back to the anchor; the lower end being supported, in the case of the first segment, by the lower hinge pin and, in the case of the next five, by the upper end of the segment next below it. This took care of twelve of the eighteen seg-

ments in each rib. The six central ones were then erected by supporting them on falsework built up on the main cables of the old bridge, which had previously been relieved of all other loads. Thus the last service of the old bridge was to assist its more massive and permanent successor into being.

Finish Is Beautiful

Considerable effort has been expended on the aesthetic treatment of the bridge and its beauty has been developed by careful treatment of its proportions and lines, rather than by addition of detailed ornamentation. There are three kinds of finish on the exposed faces. The surfaces of the gulleys were "accreted" to a true plane and then finished by stippling with a brush. The concrete surfaces were rubbed smooth with carborundum bricks after sprinkling with water; this works up a lather, which, when brushed over, sets up with a smooth even texture of a pale gray color. All panels were "brush-hammered," a process which chips the surface of the concrete, exposing the pebbles and gives a rough finish in pleasing contrast to the smoother and lighter portions of the work.

In design, unlike any other bridge yet built, constructed substantially to last for many generations; excellent in material and workmanship and moulded on bold, artistic lines, this bridge fits well the beauty of the natural setting in which it lies. With the laying out of the immediate surroundings in keeping with such a monument, this locality bids fair to become as famous for its beauty as it is for its manufactures.

LOCAL AUTO COMPANY DOUBLES ITS CAPITAL

SALEM, Or., Dec. 27.—Seven Oregon citizens and one foreign corporation transacting business in this state have filed notices of increased capital stock in the state corporation department. These increases, the corporation commissioner said, indicate that the several industries represented apparently are looking ahead to a prosperous year in 1923.

Among the increases filed was that of the Park-Shepherd Motor company of Oregon City which changed its capital stock from \$10,000 to \$20,000.

Eighteen Families Are Tendered Aid Thru Joint Relief

Eighteen needy families of Clackamas county were made happy Christmas through the efforts of the Red Cross, Improved Order of Red Men and Elks lodges of this city.

Provisions to last about two weeks' time were donated to those people, and through the kindness of other residents of this city canned fruit and clothing were donated. Miss Cis Barclay Pratt, for a number of years local secretary of the Red Cross Association, was selected to arrange the donations for distribution, and several days were required for the task. Dr. W. E. Hampstead's office was headquarters for the work.

Harry Williamson, Carl Shulson, George Chambers, Tack Stovall and Albert A. Minger donated their services in distributing the gifts throughout the city and other places in the county.

NO TIME

Oldman: One should always practice what one preaches.

Ever stop to think what would the world do if it were not for safety pins, gargles Amos Tash.

BANK OF OREGON CITY TO REMODEL BUILDING

Expenditure of \$25,000 Is Planned on Improvement Of Main Street Structure.

Contracts for the complete remodeling of the Bank of Oregon City building at Sixth and Main streets were awarded Wednesday, involving the expenditure of \$25,000 and a ninety day period of construction. The bank quarters on the main floor are to be remodeled and the office rooms on the second floor, gutted by fire September 2, are to be completely rebuilt, according to Will T. Wright, president.

The plans for the remodeling of the bank's quarters include the widening of the present office space by extension 13 feet north. This change will allow increased lobby room and added space for the offices of the bank officers. A new directors room, ladies room, room for a new department to be added later, two new tellers cages and a large new vault for the housing of the bank's records are included in the plans.

Entrance Is Improved
The entrance to the bank will be moved from the corner to practically the middle of the Main street frontage occupied by the bank proper, and the entrance to the office quarters will be shifted to the side, forming a side entrance to the bank lobby. The upper portion of the building is to embrace four new business suites of 14 rooms in addition to the room occupied by Schubel and Beatie. O. D. Eby, John F. Clark, and Will Hammond are among those who will occupy the new rooms. A special vault is being built in one of the upper offices.

Front to be Changed
The recent fire swept through the upper floor of the building, but the bank offices were not greatly damaged. The loss is set at \$5,000. Plans for the remodeling however, had been under consideration for some time.

The Bank of Oregon City owns the entire Main Street frontage extending from Sixth street to the alley in the middle of the block. The whole front of the building is to be remodeled into a modern, uniform exterior. The bank has occupied the same location since its establishment in 1881. The last remodeling was done about 12 years ago. The contract for the new work has been awarded to W. G. H. Kreuger.

O. Hall Acquitted Upon Liquor Charge

O. Hall, operator of a filling station at Gladstone, was acquitted in the justice court Wednesday of charges of illegal possession of liquor. Harry Bundell, arrested with Hall, testified that the liquor which the officers found belonged to him and Hall knew nothing of its presence. Bundell was previously sentenced to six months in jail and fined \$500 by Judge Noble when he pled guilty to the charge.

Marriage Licenses Granted 2 Couples

Marriage licenses were granted Wednesday to two couples: Alvin K. Otto, Portland and Berendence E. Hall, Oregon City; Louis Smith and Martha Aerni, both of Oswego.

LOCAL BRIEFS

Wisconsin Men Here—
Milton Knowlan and Ollie Beaver, of Green Bay, Wisconsin, who recently arrived in Oregon City, have decided to remain here until next fall. Beaver has taken a position with the Hawley Pulp & Paper Company, and Knowlan has entered the Oregon City Business College. The young men made the trip to Oregon in Knowlan's coupe. They enjoyed camp life along the route when weather conditions were favorable, and visited places of interest. Among these was Yellowstone Park, where they remained for several days. A number of interesting trips were made in the big park. High peaks were scaled and several hiking trips made. After leaving the park they camped for several days along the banks of streams, where they caught their first mountain trout. Knowlan acted as chef. They will make the return trip over the southern route, visiting California and Mexico, while enroute. During their stay in this city they are at the home of Knowlan's father, Dan Knowlan, of Fourteenth and Main streets.

Shoes Needed for Boys and Girl—
The chest for cast-off clothing always maintained at the rest room of the W. C. T. U. was completely emptied at Christmas time in order to meet the demand for clothing from needy people. A call for shoes for two boys aged 15 and 13, besides shoes for a little eight-year-old girl has just come to Mrs. E. M. Scouton, member of the W. C. T. U. The little girl was compelled to be absent from school a few days ago as her shoes were unfit for the weather. The father has been unable to secure employment, while the mother has been trying to help to make a living for the family by working in a restaurant.

R. G. Scott in Oregon City—
R. G. Scott, of Sherwood, was in Oregon City on business Wednesday. Scott served as county agent of Clackamas county for two and one-half years, resigning the position to follow farming. He is now specializing on goats and sheep of the livestock industry, while Logansberry culture and growing of filbert nuts are taken up as a side-line. The Scott farm consists of 125 acres, and is one of the best farms of that section. Richard Scott, Jr., son of Mr. and Mrs. Scott, who was taken to California for his health, is greatly improved. He is still in the south, where he is visiting friends, and will spend the remainder of the winter in California.

To Raise Fund for Home—
In order to add to the fund to be raised by Clackamas county for the Farm home at Corvallis, a home for orphans of Oregon, the local W. C. T. U. will serve lunch at the rest room at the rear of the Jones drug store today. Clackamas county's quota for the home is \$3200 and the women of the county who are members of the W. C. T. U., are working diligently to have Clackamas county go over the top in raising the fund. Mrs. Frank Welsh, who is to be hostess today at the rest room, is to be assisted by Mrs. E. M. Scouton, Mrs. N. M. Alldredge and Mrs. Ada Stevens.

Miss Shaw To Appear On Program—
Miss Virginia Shaw, student of the Washington State College, at Pullman, Wash., has returned to Oregon City for the holidays. She is visiting her parents, Mr. and Mrs. E. L. Shaw, of Sixth and High streets. Miss Shaw is among the students to appear on the program at the meeting of the Woman's Club at the Commercial Club parlors this afternoon. She will finish her course in athletics next June.

Whitman College Student Home—
Gordon Hannaford, student of Whitman College at Walla Walla, Wash., is spending the holidays with his parents, Mr. and Mrs. Charles Hannaford, of Ninth and Monroe streets. Hannaford, who graduated with honors from the Oregon City high school last June, won a scholarship entitling him to enter Whitman College. He will return to Walla Walla next Tuesday.

Dertick Club Will Meet—
Mrs. Nieta Barlow Lawrence and Mrs. William Krasig are to be hostesses to the Dertick Club Friday afternoon. The meeting will be at the home of Mrs. Lawrence on Twelfth and Washington streets, when Miss Lorraine Lee, popular vocalist of Canby, is to be the soloist.

Christmas Tree and Entertainment—
The annual Christmas tree and entertainment of St. Paul's Episcopal Sunday school will be held at the parish house at Ninth and Main streets Tuesday evening, January 2, instead of Friday evening, December 29th, as planned.

Miss Money Returns to Bend—
Miss Marjorie Money, of Bend, Oregon, who has been spending the Christmas holidays with her parents, Mr. and Mrs. William R. Money, of Gladstone, returned to Bend Tuesday morning.

Haas Spent Christmas With Mother—
Sylvester Haas, who has been enjoying the Christmas holidays with his mother, Mrs. Haas, of Greenpoint, and also visiting other relatives and friends, has returned to Reedsport. Haas is employed at that place.

Sixteen Teachers Take Examination

There were sixteen teachers of Clackamas county taking the teachers' examination in this city last Thursday and Friday. This was the smallest number of teachers in ten years to have taken the examinations.

These papers have been sent to Salem, and will be graded by the county superintendent, Brenton Vedder, when he attends the meeting of the county superintendents' state convention in that city Saturday. As examinations were conducted in every county of the state, other county superintendents will be in Salem to grade the papers of their counties.

LAMB STEW

Cut up the lamb into small pieces (after removing all the fat) say about two inches square. Wash it well and put it over the fire, with just enough cold water to cover it well, and let it heat gradually. It should stew gently until it is partly done; then add a few thin slices of salt pork, one or two onions sliced up fine, some pepper and salt if needed, and two or three raw potatoes cut up into thin pieces. Cover it closely and stew until the meat is tender. Drop in a few made dumplings, made like short biscuit, cut out very small Cook fifteen minutes longer. Thicken the gravy with a little flour moistened with milk. Serve.

COFFEE CAKE

1-2 cup butter, 1-2 cup sugar, 3 eggs, beaten, 3 tablespoons strong coffee; 1-2 cups flour; 1 teaspoon baking powder.
Cream butter and sugar together, then add eggs, beat well, also add coffee, and sift in flour and baking powder. Turn into a greased and papered tin and bake in a moderate oven for forty minutes. When cold cover with boiled frosting.

Napoleon Ingalls Old River Captain Dies in Portland

Napoleon Bonaparte Ingalls, one of the old time rivermen, died Tuesday at the Mann home in Portland. He had been a resident of Portland since 1853.

Ingalls was one of the first pursers on the old steamers Belle and Eagle on the Oregon City run, in 1853. He was later on the river steamers Portland, Jennie Clark, Rival and Express, being on the run until 1858, when he went on the route to the Cascades, remaining until 1893. In that service he was on the steamers Carrie Ladd, Montana Buck, Julia, Cascade, Wilson G. Hunt, New World, Oneonta, Dixie Thompson, Emma Hayward, Wide West, S. G. Reed, Bonita, R. R. Thompson, Multnomah, Hassalo, Lurline and Astorian.

Ingalls was born in Sandburton, New Hampshire, December 11, 1830. He lived there until 11 years of age, when he went to Hartford, Conn. After two years he went to Boston and left there for California in July, 1852, by way of the isthmus of Panama, coming to Portland a year later.

He was married in Oregon City in 1855. Mrs. Mabel Jones, a daughter, is living at Oakland, Cal., and Percy Ingalls, a son, is at Anaconda, Mont.

MILK AS FOOD

In the preservation of foods, an art which has reached considerable perfection in these days, it is interesting to note how important a part is played by opposite forces of nature.

For instance, after milk is obtained from the cow, it is necessary to reduce it to low temperature so as to inhibit the growth of bacteria.

Later, however, an opposite plan is essential. By prolonged heating of the milk in a moderately high temperature the bacteria or germs which are unhealthful are killed and the milk, unchanged in its digestibility, is rendered safe for use by the infant or invalid.

Both of these methods are used in their simplest form in the preparation of malted milk, so as to obtain the highest purity of the food.

The absence of moisture is an efficient safeguard against the growth of bacteria and it also makes the product capable of being taken easily to all parts of the world.

It was no little achievement to make a product which retains all the nutritive value of malted grain and rich milk and which produces at a moment's notice a food-drink that is sustaining, invigorating and also delicious to the taste.

POTTED HAM

To two pounds of lean ham allow one pound of fat, two teaspoonsful of powdered mace, half a nutmeg, grated, rather more than half a teaspoonful of cayenne.

Mode—Mince the ham, fat and lean together, in the above proportion, and pound it well in a mortar, seasoning it with cayenne pepper, pounded mace and nutmeg; put the mixture into a deep baking dish, and bake for half an hour; then press it well into a stone jar, fill up the jar with clarified lard, cover it closely, and paste over it a piece of thick paper. If well seasoned, it will keep a long time in winter, and will be found very convenient for sandwiches, etc.

Combine one cupful of very strong coffee with two cupfuls of light cream, one and an eighth cupfuls of maple syrup, a few grains of salt, an egg yolk and one and a half teaspoonsful of vanilla. Freeze in an ice cream freezer, in three parts of cracked ice to one of rock salt.

COUNTY FARM BUREAU TO HOLD ANNUAL MEET

Officers for Coming Year to Be Elected on January 2; New Projects to be Talked.

The annual meeting of the Clackamas County Farm Bureau will be held here Tuesday, January 2, according to the announcement of Mrs. John Caffrey, secretary-treasurer of the organization. The session which will be attended by farmers from over the entire county is to be held at 10 o'clock in Willamette hall.

Election of officers, appointment of the executive committee and the outlining of the agricultural improvement projects to be undertaken by the organization during the coming year, will occupy the session. The coming meeting of the legislature will be the subject of some discussion, together with the determination of what measures of an agricultural bearing nature the organization will support.

Will Deer Suicide? Molalla Man Says That It Isn't Done

Controversy over the question of animal suicide has arisen since the publication of a story concerning the supposed "suicide" of a buck deer. Writing to the Oregonian from Molalla, J. W. Thomas, generalizes upon the points covered in the story of the buck deer who committed suicide off of a 50-foot cliff near Estacada. Thomas writes:

"In the Oregonian we notice the account of the robust buck deer committing suicide off a 50-foot cliff at Estacada. The inference from this account of the deer's death near the dam site is that the deer deliberately made tracks up to the cliff's edge and purposely jumped off to take his life. Old hunters would call this bunk by the dam site.

"Having had experience in following deer on their run over steep and dangerous places when dry, it is no wonder to me that deer sometimes fall to their death when crowded over dangerous runs slippery with ice.

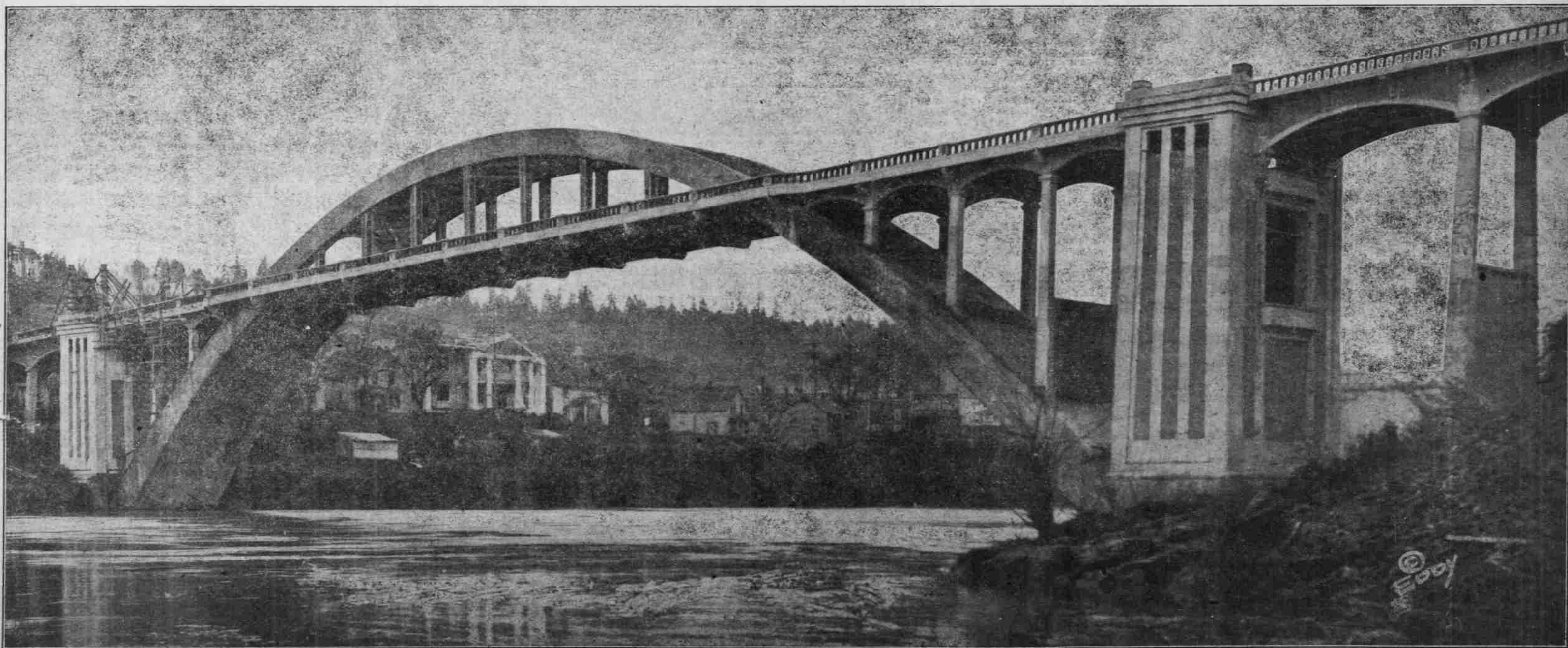
"Should a believer in the so-called deer suicide take trouble to back track the deer far enough from the cliff, the cause will surely come to observation. He may, in a measure be in the deer's place and as he tumbles to the situation, and in his rush even to climb down a ladder, his grip may fall on an icy rung and he, too, fall to his accidental death."

Local Delegates Go To Teachers' Meeting

Delegates attending the State Teachers' Annual convention now in session in Portland, who are representing Clackamas county are County Superintendent Brenton Vedder; R. W. Kirk, city superintendent; Mrs. Ethel Lowndowne, county supervisor; J. W. Leonard, principal of Gladstone school; Louise Reese, principal of Harmony school; Miss Bernice Cunningham, of Concord; Edgar A. Means, principal of the Colton school. Superintendent Vedder will go to Salem Saturday, where he will attend the county superintendents' convention.

"How do you meet your expenses?" "Going and coming."

"THE MOST BEAUTIFUL BRIDGE IN AMERICA," SPANNING THE WILLAMETTE BETWEEN OREGON CITY AND WEST LINN



The new arch, upon the Pacific Highway between Oregon City and West Linn, was designed by the state highway commission bridge engineer C. B. McCullough, and its construction supervised by Engineers R. A. Furrow and C. P. Richards. It was built by A. Guthrie and Company, Inc., of Portland. Its total length is 850 feet. Deck above low water at mid-span, 77 ft. Main span: steel rib arch with gunite encasement; 350 ft. span, 100 ft. high. Approaches: reinforced concrete beam and slab spans, 35 ft. centers. Height of West Linn pier, rock foundation to road level, 98 ft. Road: slope, 5 per cent; width, 13 1-2 ft. Two sidewalks, 4 3-4 ft. each. Weight of steel in arch, 480 tons; reinforcing steel, 150 tons. Volume, concrete, 4600 cu. yds. Volume, gunite, 250 cu. yds. Construction started July 29, 1921. Completed December, 1922. Cost \$275,000.00.

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